

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 10 - Formative Assessment 1

Formative Assessment (FA-1)

Class: Class 10

Assessment Type: Formative Assessment (FA-1)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Networking Concepts, Internet and Web Technologies

ICT Teaching Resources Portal

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

- a. Which of the following is NOT a type of computer network?
 - b. LAN
 - c. WAN
 - d. MAN
 - e. CAN
- b. The full form of TCP/IP is:
 - c. Transfer Control Protocol/Internet Protocol
 - d. Transmission Control Protocol/Internet Protocol
 - e. Transmission Communication Protocol/Internet Protocol
 - f. Transfer Communication Protocol/Internetwork Protocol
- c. Which device is used to connect multiple networks and direct data packets?
 - d. Hub
 - e. Switch
 - f. Router
 - g. Modem
- d. The top-level domain .edu is used for:
 - e. Government websites
 - f. Educational institutions
 - g. Commercial organisations
 - h. Network infrastructure
- e. What is the maximum number of characters allowed in a URL as per standard practice?
 - f. 2048
 - g. 255
 - h. 1024
 - i. 512

Section II: Short Answer (5 x 2 = 10 marks)

- a. Differentiate between **LAN** and **WAN** with two examples each.
- b. Explain the client-server model with a neat labelled diagram.
- c. What is the role of a DNS server? Explain how domain name resolution works.
- d. Compare HTTP and HTTPS. Why is HTTPS preferred for banking websites?
- e. What is cloud computing? List any two service models (IaaS, PaaS, SaaS).

Section III: Long Answer (5 x 1 = 5 marks)

- a. Explain the TCP/IP protocol suite with the help of a diagram. Describe the function of each layer and give one protocol used at each layer. **(Board Exam Pattern - 2024)**

Part B: Hands-on Practical Lab Task (5 marks)

Write a Python program that:

- Reads a CSV file containing student data (Name, Roll, Marks in 3 subjects)
- Calculates total and percentage for each student
- Writes the results to a new file with Grade column (≥ 90 : A, ≥ 75 : B, ≥ 60 : C, < 60 : D)
- Displays the topper of the class

Part C: Notebook Maintenance (5 marks)

- a. Neatness and organisation of notebook (2 marks)
- b. Completion of all assigned classwork and homework (2 marks)
- c. Proper headings, dates, and indexing (1 mark)